Running Evo 2 7B on Google Cloud and Runpod: cost, GPU memory and cross-hardware agreement of variant scores

Study repository `rewire-bio/evo2-cloud-operations`

October 2026

Abstract

Clinical informatics teams evaluating the Evo 2 DNA language model need to know what it costs to run, what hardware it needs, and whether its outputs are stable when the hardware changes. We ran one pinned Evo 2 7B container image, unchanged, on three single-GPU platforms: a Google Cloud `a3-highgpu-1g` Spot VM (H100 80 GB), a Runpod Secure Cloud H100 80 GB pod and a Runpod L40S 48 GB pod. Each platform ran Arc Institute's forward-pass correctness test, zero-shot scoring of 3,893 BRCA1 single-nucleotide variants with 8,192 bp windows, embedding extraction for an exon classifier, a memory probe over sequence length and a same-machine repeat. All platforms passed the correctness test (loss 0.34766). The two H100s, from different providers, produced bit-identical variant scores and classifier probabilities. The L40S produced different scores: median absolute difference 1.2e-04 against the H100, Spearman correlation 0.983, and 2,627 of 3,893 variants moved by more than 1% of the list in rank, while the AUROC for loss-of-function variants changed only slightly (0.877 against 0.875). Every machine reproduced its own scores exactly. One forward pass fitted at 32,768 bp and ran out of memory at 131,072 bp on both the 48 GB and the 80 GB GPU; lengths in between were not tested. Scoring cost USD 0.52 (Runpod H100), 0.68 (Runpod L40S, which varied with the host) and 0.80 (Google Cloud H100 Spot) per 1,000 variants at list prices. Zero-shot Evo 2 scores separated BRCA1 loss-of-function variants better than the CADD v1.3 scores supplied with the BRCA1 table (AUROC 0.875 against 0.804); the difference was concentrated in noncoding variants, and on coding variants no difference was detected. Against current CADD (v1.7) and SpliceAI, looked up for the same variants afterwards, no noncoding difference was detected, and current CADD and REVEL were better on coding and missense variants. At decision thresholds fixed on the H100, 1.8% to 4.9% of variants changed call on the L40S; computing in FP32 removed almost all of these changes. These are operational measurements, not a clinical validation.

1 Introduction

Evo 2 is an open DNA language model with 7B and 40B parameter checkpoints and a context of up to 1 Mb [1]. Its authors report strong zero-shot variant-effect results, particularly for noncoding and splice variants, and publish inference code, weights and notebooks [2]. They do not describe how to provision it on commercial clouds, what it costs, or how its outputs behave across GPU types. Users have reported that 7B embeddings differ between H100 and L40S GPUs [3] and that long sequences exhaust GPU memory well short of the advertised context [4]. For a clinical team these are not curiosities: a score that changes with the hardware cannot be locked down by an assay validation, and a context that does not fit is not a usable context.

This study runs one fixed workload on several platforms with the same container image, and records correctness, agreement, memory, time and cost. It also re-implements two of Arc's published notebook workflows (BRCA1 zero-shot scoring, which Arc demonstrates with the 1B checkpoint, and exon classification) with the 7B checkpoints, against labels and simple baselines; no published 7B reference value was compared. The design, platforms and budget were approved in a written protocol before any GPU run, with four dated amendments (amendment 4 records

deviations from the original run plan, acknowledged by the study owner). Amendments 5 to 7, each written before its analysis, add follow-up analyses (Section 5).

2 Methods

2.1 Platforms

Table 1 lists the platforms. All are single-GPU machines. Google Cloud's `a3-highgpu-1g` is sold only as Spot or Flex-start; we used Spot. Runpod pods ran in Secure Cloud, on demand. The Google Cloud H100 is the reference platform for agreement comparisons. AWS `p5.4xlarge` and `g6e.2xlarge` launchers are included in the repository but were not run (no account was available; amendment 2). Runpod chose the data centre for each pod; we did not constrain it.

2.2 Software and model pins

The container is built from `nvcr.io/nvidia/pytorch:25.04-py3` (the base of Arc's Dockerfile), pinned by digest, with `evo2==0.6.0` and `vtx==1.1.0`. It contains PyTorch 2.7.0 (NVIDIA build), flash-attn 2.7.3 and Transformer Engine (`container/pip-freeze-22970720.txt`). Because Transformer Engine is present, the 7B checkpoints ran with their configured FP8 input projections (`use_fp8_input_projections: True`) on every platform, not in pure bfloat16. The image is referenced by digest on every platform. Weights were downloaded from Hugging Face at pinned revisions and checked against SHA-256 hashes before loading, because Arc's loader uses `torch.load`, which can execute code. The exon classifier (`schmojo/evo2-exon-classifier`) is a two-layer network; instead of loading it with `trust_remote_code` as Arc's notebook does, we rebuilt the network locally and loaded only the pinned `safetensors` weights. Study code was copied to each machine from a committed revision; all three platforms in the recorded run used the same revision.

2.3 Data

All inputs are public and were fetched from Arc's repository at a pinned commit, with SHA-256 checks: GRCh37 chromosome 17; the Findlay et al. BRCA1 saturation genome editing table [5] (3,893 SNVs: 2,821 functional, 249 intermediate, 823 loss of function, with consequence, CADD v1.3 and mammalian phyloP columns, as annotated by Findlay et al.); its function scores are freely available for nonprofit use, and this study is a nonprofit use; 122 labelled exon and non-exon positions; and Arc's four forward-pass test sequences. No patient data were used.

2.4 Workloads

W0 reimplements Arc's forward-pass test with `evo2_7b`; a platform passes if the mean loss is within 0.001 of Arc's expected 0.3477. **W1** scores 8,192 bp windows centred on each BRCA1 SNV, as in Arc's notebook: the score is the variant window's mean log-likelihood minus the reference window's (batch size 1). The 3,893 variants share 1,326 reference windows, so 5,219 sequences are scored. **W2** extracts final-token embeddings from layer `blocks.26` of `evo2_7b_base` for forward and reverse 8,192 bp sequences at 122 positions and scores them with the pinned classifier. **W3** runs one forward pass of `evo2_7b` on synthetic sequences of 8,192, 32,768, 131,072 and 262,144 bp and stops at the first out-of-memory error. **W4** rescores the first 200 BRCA1 variants in a new process on the same machine.

2.5 Metrics

Discrimination is the AUROC for loss-of-function against functional or intermediate variants, using the negated W1 score, with 95% intervals from 2,000 bootstrap resamples of positions

(the three SNVs at a position are kept together). Baselines from the same table need no GPU: CADD v1.3, mammalian phyloP (amendment 1 corrected its sign before any Evo 2 output existed) and a consequence rule (nonsense and canonical splice, then splice region, then missense, then other). Paired differences use the same resamples. Agreement between platforms is the maximum and median absolute difference in per-variant scores, the Spearman correlation, and the number of variants whose rank moves by more than 1% of the list (39 places). Peak memory is `torch.cuda.max_memory_allocated`, with `nvidia-smi` sampled every 500 ms. Cost is billed time (from pod creation on Runpod, from `RUNNING` on Google Cloud, to deletion) times the list price retrieved by API at run time, plus disk.

3 Results

3.1 Correctness and agreement

All three platforms passed W0 with mean loss 0.34766. The gate has bfloat16 resolution: per-sequence losses were identical on all GPUs, but token accuracy differed on the L40S (0.8626 against 0.8636 in the recorded run), so W0 cannot detect differences of the size seen in W1. Each platform reproduced its own W4 scores exactly. The Runpod H100 matched the Google Cloud H100 exactly on all 3,893 variant scores and all 122 classifier probabilities (maximum absolute difference 0). The L40S did not (Figure 1): its scores differed from the reference by a median of 1.2e-04 and at most 1.3e-03, with Spearman correlation 0.983; 2,627 variants moved by more than 39 rank places, and classifier probabilities differed by up to 0.020. For scale, the BRCA1 variant scores themselves are of order 10^{-4} to 10^{-2}. Two earlier runs and the independent clean reproduction, each on fresh machines on the same day, gave BRCA1 scores and exon probabilities bit-identical to the recorded run on every platform, including L40S hosts in different data centres with different drivers and kernels (`evidence/checks/verification.json`). One of those runs, 698234c1, was rejected by the harness because the coordinator changed analysis code while it ran; its outputs are used here only for this comparison. Among loss-of-function variants alone, the L40S and H100 rankings correlated more strongly (Spearman 0.998), and 48 of the 50 lowest-scoring variants were shared. The L40S AUROC was higher by 0.002 (95% interval 0.0002 to 0.004, paired grouped bootstrap), a small but detectable difference (`evidence/checks/verification.json`).

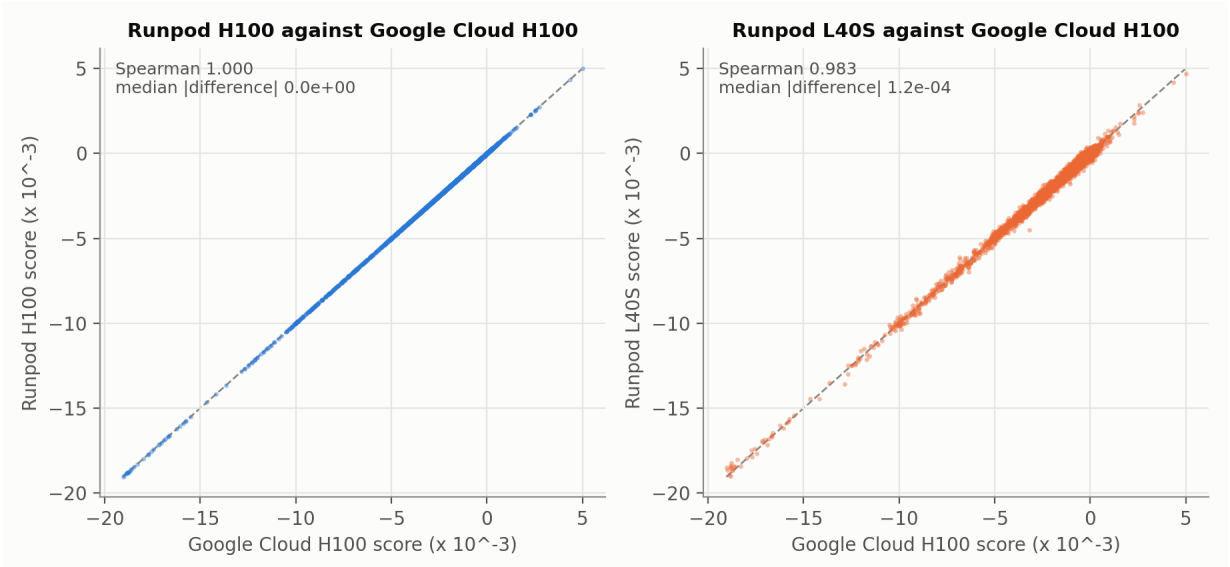


Figure 1: Per-variant BRCA1 scores on each platform against the Google Cloud H100. Same image, inputs and code. Left: Runpod H100 (identical). Right: Runpod L40S. Dashed line: identity.

3.2 BRCA1 discrimination and baselines

Zero-shot Evo 2 7B scores gave AUROC 0.875 [0.859, 0.891] for all SNVs, against 0.804 for CADD v1.3, 0.781 for phyloP and 0.769 for the consequence rule (Figure 2). Evo 2 exceeded CADD v1.3 by +0.071 [+0.050, +0.095]. Within subsets, the difference was concentrated in noncoding variants (Evo 2 0.953 against CADD 0.875; difference +0.077 [+0.044, +0.113]); on coding variants no difference was detected (Evo 2 0.840, CADD 0.841; difference -0.001 [-0.026, +0.023]), which does not establish that they are equal. AUROCs for subsets do not add up to the overall AUROC, which also compares variants across subsets. The CADD scores are version 1.3, as supplied in the Findlay et al. table; later CADD releases added splicing predictors, which would be expected to help most on exactly the noncoding and splice-region variants where Evo 2 led; Section 5.3 compares them. Intervals resample positions; neighbouring positions share almost all of their 8 kb window, so the intervals may be too narrow. The exon classifier gave AUROC 0.945 on 122 positions; its training data may include these positions, so this is a pipeline check rather than an evaluation.

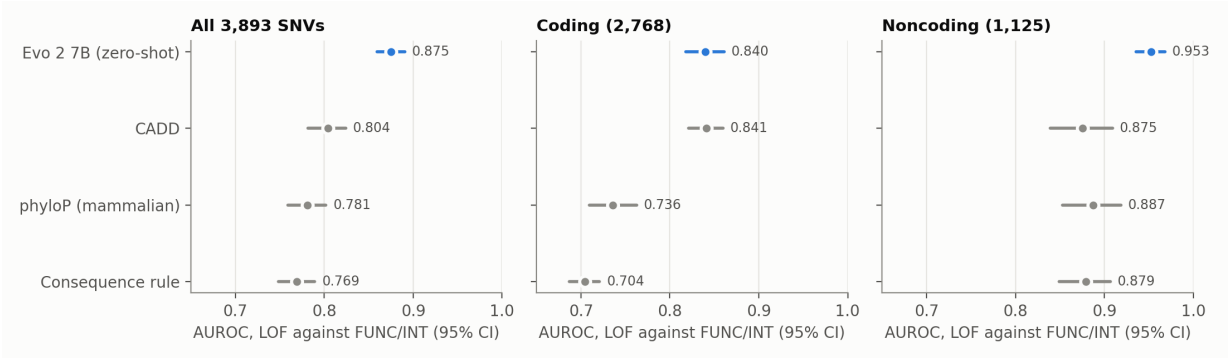


Figure 2: AUROC for BRCA1 loss-of-function variants with 95% grouped bootstrap intervals, reference platform. Baseline scores (CADD v1.3, mammalian phyloP) come from the Findlay et al. table.

3.3 Memory

Scoring 8,192 bp windows needed 16.8 GiB allocated by PyTorch on every platform. A single 32,768 bp forward pass needed 30.4 GiB. At 131,072 bp the forward pass ran out of memory on both the 80 GB H100 and the 48 GB L40S (Figure 3); lengths between 32,768 and 131,072 bp were not tested. With this code path (one full forward pass returning all logits), the checkpoint's 1 Mb context was not reachable on either GPU; Arc points to multi-GPU frameworks or slower forced prompting for long sequences [4].

3.4 Throughput and cost

Table 1: Platforms and measured operations for the recorded run. USD per 1,000 variants assumes the BRCA1 pattern of 5,219 scored windows per 3,893 variants and list prices at run time.

Platform	GPU	Location	Purchase	USD/h	Windows/s	USD/1k var.	Minutes	USD
Google Cloud H100	H100	us-central1-a	spot	6.29	2.93	0.80	46	4.78
Runpod H100	H100	US-MO-1	on-demand	4.02	2.91	0.52	42	2.82
Runpod L40S	L40S	US-MO-1	on-demand	1.12	0.61	0.68	159	2.96

The two H100s scored at 2.93 and 2.91 windows per second and the L40S at 0.61 (Table 1). L40S throughput depended on the host: L40S pods in Runpod's US-TX-4 data centre scored 0.92 to 0.93 windows per second in the phased and rejected runs, against about 0.6 on the US-MO-1 hosts used in the recorded run (run log). Its cost per variant therefore ranged from below to above

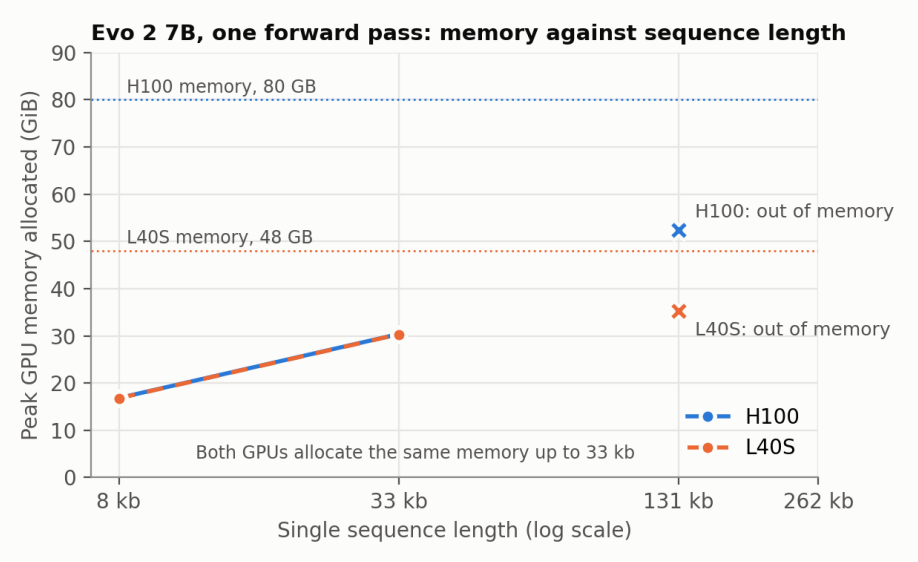


Figure 3: Peak GPU memory for one forward pass against sequence length. Crosses mark the memory in use when the next allocation failed. Dotted lines show nominal capacity (80 and 48 GB); PyTorch reported 79.2 and 44.4 GiB usable.

that of the Runpod H100, depending on the host it was given. The Google Cloud Spot H100 cost more per variant than the Runpod on-demand H100, because its hourly price was higher at the same throughput; Spot capacity can also be reclaimed mid-run, which did not happen here. Setup took a large share of this short workload: pulling the 12 GB compressed image (12.11 GB of base layers, per the registry manifest) took 491 s on Google Cloud in the recorded run, and Google's current Deep Learning VM image needed Docker and the NVIDIA container toolkit installed first (21 s in the recorded run, 457 s in the phased run). The completed attempts in Table 1 cost USD 10.56; including the failed L40S attempt (USD 0.82), the recorded run cost USD 11.38 (`evidence/checks/verification.json`). The whole study, including phased, rejected and reproduction runs, cost about USD 49 in cloud compute: USD 21.56 on Runpod (from its billing API) and about USD 27 on Google Cloud (billed VM time at the Spot list price); Google Cloud billing was not reconciled line by line.

3.5 Operational events

The run log records every attempt. Earlier, phased runs on the same three platforms found and fixed four launcher problems before the recorded run: Runpod's REST API v1 had been superseded by v2; a Runpod account without credit was refused with HTTP 402; Google's CUDA 12.9 Deep Learning VM image ships without Docker; and it holds the NVIDIA container toolkit at an older version, so the current package cannot be installed alongside it. Runpod placed one H100 pod in India and the L40S in Texas, because no data centre was specified. In the rejected run 698234c1, both Google Cloud attempts lost GPU access for new processes inside the running container after the main workloads had finished, probably because a systemd reload on the host revoked the container's device permissions. For the recorded run, the launcher attached the GPU through CDI and paused automatic updates, and every workload completed. In the recorded run, the first L40S attempt failed when the SSH session from the coordinating laptop dropped, because the workload ran in the foreground of that session; the second attempt completed. The run plan itself deviated from the approved protocol (more attempts in the phased runs, and several runs rather than one); amendment 4 lists the deviations.

4 Discussion

Hardware is part of the method. With the same image, inputs and GPU model, Evo 2 7B outputs reproduced exactly across two providers and across processes. Changing the GPU model changed individual scores by about as much as the typical (median) score, reordering most of a ranked list dominated by near-zero scores, while aggregate discrimination changed by only 0.002 in AUROC. Reference-window log-likelihoods themselves shifted by about 10^{-4}, which is enough to flip the sign of small scores. The most damaging variants were more stable: 48 of the 50 lowest-scoring variants were shared between the GPUs. The 7B checkpoint uses FP8 input projections when Transformer Engine is present, and FP8 and bf16 kernels differ between GPU generations, which is a plausible cause; the follow-up (Section 5) found it to be a partial one. In practice a validated pipeline must fix the GPU model or revalidate when it changes, and Arc's correctness test is not sufficient on its own: the L40S passed it.

Context capacity is not memory fit. On both GPUs tested, a single forward pass fitted at 32 kb and failed at 131 kb; extrapolating the measured memory linearly suggests that about 65 kb might fit on an 80 GB H100 but not on a 48 GB L40S, which was not tested. Pipelines that rely on long context, such as EVEE's 65 kb windows over four million variants (about 20,000 H100-hours) [6], are a different operational class from the 8 kb scoring measured here.

Price and residency. A cheaper 48 GB GPU is adequate for 8 kb scoring, but on a GPU cloud its cost per variant depends on the host it lands on, so measure throughput before committing to a large job. Spot pricing is not automatically cheaper than a GPU cloud's on-demand price. GPU clouds choose a location unless told otherwise; with identifiable data the location must be pinned, and covered by a business associate agreement where HIPAA applies. EC2, EBS and Compute Engine are covered by their providers' BAAs [7, 8]; Runpod states HIPAA and GDPR programmes [9]. NVIDIA's hosted Evo 2 API prohibits protected health information and production use [10].

5 Follow-up analyses

5.1 Cause and control (amendment 5)

A follow-up on 11 October 2026 scored every 8th BRCA1 variant (487) on a Runpod H100, L40S and L40 under three numerical configurations; controls reproduced the recorded run exactly. Turning off the FP8 input projections reduced the median H100 against L40S difference from 1.2×10^{-4} to 8.4×10^{-5}, so FP8 contributes but is not the main cause. Computing in FP32 (TF32 disabled, PyTorch attention) reduced it to 4.8×10^{-7} (maximum 2.9×10^{-6}), with identical rankings, at 5.8 times the H100 time and 3.4 times the L40S time and 29.6 GiB of memory. An L40 and an L40S, both Ada Lovelace, disagreed as much as the H100 and L40S did, so validation must be per GPU product, not per architecture. Scoring in batches of 8 was bit-identical to batch size 1 (`followup/RESULTS.md`).

5.2 Changes of call at a fixed threshold (amendment 6)

Rank changes do not show how many variants would be classified differently. Using stored scores only, we set three thresholds on the Google Cloud H100 scores (Youden, and sensitivity 0.90 and 0.95 for loss of function; Arc's notebook defines none) and applied them unchanged to the L40S. At the Youden threshold 72 of 3,893 variants (1.8%; exact 95% interval 1.5% to 2.3%) changed call; at sensitivity 0.90 and 0.95, 152 (3.9%) and 189 (4.9%). Changes ran in both directions in similar numbers, and half of the changed variants lay within 0.7 median cross-GPU differences (about 8×10^{-5}) of the threshold. Controls (H100 against H100, and each platform's recorded run

against its reproduction) gave no changes. On the 487-variant follow-up subset, with thresholds re-derived on the H100, 8, 13 and 22 variants changed call in the default configuration and 0, 0 and 1 in FP32; an L40 against an L40S gave 8, 14 and 25 (`followup/CLASSIFICATION-FLIPS.md`).

5.3 Current predictors (amendment 7)

We looked up current predictors for the same 3,893 variants from published precomputed tables: CADD v1.7 [11], SpliceAI [12] (Illumina's masked precomputed scores, as annotated in the CADD v1.7 release) and, for missense variants, REVEL v1.3 [13]. Every variant had CADD v1.7 and SpliceAI scores and every missense variant a REVEL score. With the W1 method (Table 2), Evo 2's advantage over CADD v1.3 on noncoding variants was not detected against CADD v1.7 (difference 0.021, 95% interval −0.003 to 0.045) or SpliceAI (−0.011, −0.025 to 0.003). On coding variants CADD v1.7 was better than Evo 2 (−0.052, −0.073 to −0.030), and on missense variants so were CADD v1.7 (−0.042) and REVEL (−0.081, −0.114 to −0.048). Of the 239 noncoding loss-of-function variants, 230 are canonical splice or splice-region variants, which is where SpliceAI is designed to work. CADD v1.7 uses SpliceAI as a feature, so the two are not independent, and REVEL's training data may include BRCA1 missense variants (`followup/CURRENT-PREDICTORS.md`).

Table 2: AUROC for BRCA1 loss-of-function variants, Evo 2 7B (Google Cloud H100) against current predictors on the same variants, computed as in Figure 2. Intervals and paired differences are in `followup/CURRENT-PREDICTORS.md`.

	All (3,893)	Coding (2,768)	Noncoding (1,125)	Missense (2,086)
Evo 2 7B	0.875	0.840	0.953	0.785
CADD v1.3	0.804	0.841	0.875	0.738
CADD v1.7	0.881	0.892	0.932	0.828
SpliceAI	0.694	0.588	0.963	–
REVEL v1.3	–	–	–	0.867

6 Limitations

- Few runs per platform; timings are receipts, not benchmarks, and throughput varied between hosts with the same GPU model. List prices change.
- Three platforms, two GPU models. AWS was not run. Blackwell GPUs, where a user reports wrong losses [14], were not tested.
- Only the 7B checkpoints, run with FP8 input projections. The 40B model needs two H100s or one H200 [15].
- The main CADD baseline is v1.3, from the BRCA1 table. Current predictors were compared by lookup only (amendment 7); REVEL may be optimistic on BRCA1 missense variants.
- The decision thresholds in amendment 6 were set on the same scores they were tested on, so they measure how often calls change, not the performance of a validated threshold.
- The BRCA1 and exon workflows are re-implementations of Arc's notebooks with the 7B checkpoints; no published 7B reference value was compared.
- The BRCA1 comparison is one gene from one assay; the exon task may overlap the classifier's training data. Neither says anything about clinical validity, calibration or ancestry.

- The follow-up identified low-precision rounding as the cause in aggregate (FP32 removed the difference) but did not isolate which kernels differ between GPU products.
- Compliance terms were taken from provider pages, not audited.

7 Availability

Code, protocol, amendments, run log, results and this manuscript: https://github.com/rewire-bio/evo2-cloud-operations. Container image: `ghcr.io/rewire-bio/evo2-cloud-operations` (digest in `configs/full.json`).

References

[1] Garyk Brixi, Matthew G. Durrant, Jerome Ku, et al. Genome modelling and design across all domains of life with Evo 2. *Nature*, 2026. doi: 10.1038/s41586-026-10176-5.

[2] ArcInstitute/evo2, commit 53f1959. https://github.com/ArcInstitute/evo2/tree/53f195997257c56c00e5ef8d33a54f5baad143a6, 2026. Retrieved 7 October 2026.

[3] Evo2 7B embeddings on H100 vs L40S, issues 177 and 178. https://github.com/ArcInstitute/evo2/issues/178, . Retrieved 7 October 2026.

[4] Embeddings on medium sized sequences (50,000 bases) cause OOM, issue 160. https://github.com/ArcInstitute/evo2/issues/160, . Retrieved 7 October 2026.

[5] Gregory M. Findlay, Riza M. Daza, Beth Martin, et al. Accurate classification of BRCA1 variants with saturation genome editing. *Nature*, 562:217–222, 2018. doi: 10.1038/s41586-018-0461-z.

[6] Michael T. Pearce, Thomas Dooms, et al. EVEE: Interpretable variant effect prediction from genomic foundation model embeddings. bioRxiv, https://www.biorxiv.org/content/10.64898/2026.04.10.717844v2.full, 2026.

[7] Amazon Web Services. HIPAA eligible services reference. https://aws.amazon.com/compliance/hipaa-eligible-services-reference/. Retrieved 7 October 2026.

[8] Google Cloud. HIPAA compliance on Google Cloud. https://docs.cloud.google.com/docs/security/compliance/hipaa. Retrieved 7 October 2026.

[9] Runpod. Compliance and certifications. https://www.runpod.io/legal/compliance. Retrieved 7 October 2026.

[10] NVIDIA. NVIDIA API trial terms of service. https://assets.ngc.nvidia.com/products/api-catalog/legal/NVIDIA%20API%20Trial%20Terms%20of%20Service.pdf, . Retrieved 7 October 2026.

[11] Max Schubach, Thorben Maass, Lusiné Nazaretyan, et al. CADD v1.7: using protein language models, regulatory CNNs and other nucleotide-level scores to improve genome-wide variant predictions. *Nucleic Acids Research*, 52:D1143–D1154, 2024. doi: 10.1093/nar/gkad989.

[12] Kishore Jaganathan, Sofia Kyriazopoulou Panagiotopoulou, Jeremy F. McRae, et al. Predicting splicing from primary sequence with deep learning. *Cell*, 176:535–548, 2019. doi: 10.1016/j.cell.2018.12.015.

[13] Nilah M. Ioannidis, Joseph H. Rothstein, Vikas Pejaver, et al. REVEL: an ensemble method for predicting the pathogenicity of rare missense variants. *American Journal of Human Genetics*, 99:877–885, 2016. doi: 10.1016/j.ajhg.2016.08.016.

[14] Unexpected loss on Blackwell, issue 157. https://github.com/ArcInstitute/evo2/issues/157, . Retrieved 7 October 2026.

[15] NVIDIA. Evo 2 NIM: prerequisites and support matrix. https://docs.nvidia.com/nim/bionemo/evo2/latest/prerequisites.html, . Retrieved 7 October 2026.